25901016

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU – BVOC (HONOURS) Regular Examination November 2025

First semester

Skill Development Components (SDC)-MG1SDCAFP102

Dairy Technology

Time: 1 hr 30 min Marks: 50

PART -A

Answer any 5 questions

(Each questions carries 2 marks)

1. What do you mean by platform test? Name the different platform tests?
   platform test, often referred to as **cross-platform testing** or simply **compatibility testing**, is a type of non-functional software testing designed to ensure that a software application or system performs correctly and consistently across different operating environments
   Smell, Appearance, Temperature, Sediments Lactometer reading etc
2. Explain about the colour defects in icecream

   Unnatural defect, using wrong colour, prevent by using proper colour

3. List the criteria for grading cheese and write the uses of cheese?
   Cheese grading criteria focus on sensory evaluation across **Flavor, Body & Texture, Color, and Finish & Appearance**, assessing how well a cheese meets its specific type's standards, with judges looking for desirable traits like balanced taste, smooth paste, uniform color, and correct eye formation (for Swiss) while deducting points for defects like off-flavors, grittiness, or discoloration.

4. List any four ways of microbial contamination in milk?
   Souring, Souring and gassiness, Aroma Production, Ropiness

5. List the different types of fermented milk?
   Fermented milk products are diverse, including global favourites like **yogurt, kefir, and buttermilk**
6. Explain PFA? Give the PFA standards of cow and buffalo milk
   Prevention of food adulteration Act

7. Pasteurization process is not a sterilization process .Explain the statement

Sterilization is a drastic process that involves heating food or objects to temperatures **above 100°C (212°F)**, often using an autoclave (steam under pressure). This intense heat is designed to destroy all microorganisms and their resilient **bacterial spores**. Products are rendered completely sterile and can be stored at room temperature for a long duration, often many months, until the package is opened.

Pasteurization uses milder heat treatments, typically **below 100°C**, for a specific duration (e.g., heating milk to 72°C for 15 seconds). The objective is to target and destroy pathogenic bacteria and inactivate spoilage enzymes, making the food safe for consumption and extending its shelf life by a few weeks. However, the less resistant forms and some heat-resistant spores can survive the process.

PART -B
(Answer 1 questions)
Each question carries 5 marks

8. How irradiation affect the nutritive value of milk?

Irradiation significantly improves milk safety by killing microbes but causes minor nutritional changes, primarily affecting some vitamins (like B vitamins) and amino acids (like tryptophan, glutamic acid), similar to heat pasteurization, with effects depending heavily on the dose; however, overall protein digestibility and biological value remain largely unaffected or slightly reduced, and with proper gassing (nitrogen), lipid peroxidation is minimized, ensuring it remains safe and nutritious, comparable to heat-treated milk

9. Write a short note on the classification and production of icecream?

Plain, Chocolate, Fruit, Nut, Ices, Sherbet, Fancy Moulded, Novelties, Soft ice-creams

Selection of ingredients
Figuring of mix
Making the mix
Pasteurizing the mix
Homogenization
Cooling and ageing
Freezing the mix
Packaging of icecream

Hardening and storage of icecream loop

10. Explain about the physico chemical properties of minor constituents?

Phopholipids, Cholesterol, Pigments, Enzymes, Vitamins

11. Explain about the direct and desi cream method of manufacturing of ghee?

The **desi method** (also known as the Bilona method) is a traditional, labor-intensive process that uses cultured curd as its base, resulting in highly aromatic and nutritious ghee. The **direct cream method**, in contrast, is a faster, more modern process that heats collected milk cream directly, which is more efficient for commercial production but typically yields a milder flavor.

12.Write a note on toned milk processing

Toned milk is made by combining whole milk (like buffalo or cow's milk) with skim milk or skim milk powder and water. This mixture is then standardized to a specific fat content, typically around 3%, while maintaining essential nutrients like calcium and protein. The process reduces the fat content of the milk, making it lower in calories and a popular alternative to full-fat milk.

13. Write down the steps involved in the pasteurization of milk?

Pasteurization of milk is heating it to a specific temperature for a set time to kill harmful bacteria like *Salmonella*, *E. coli*, and *Listeria*, making it safe to drink and extending its shelf life, without significantly reducing nutrients. Common methods include **HTST (High-Temperature Short-Time)**, heating to ~72°C (161°F) for 15 seconds, and **UHT (Ultra-High Temperature)**, using much higher temps for shorter times for longer shelf stability. This process is crucial for public health, preventing diseases like tuberculosis and typhoid, and ensures milk remains a safe, nutrient-rich food.

- Receiving of milk
- Preheating
- Filtation
- Cooling to 5 degree
- Standardizing
- Pasteurization
- Cooling
- Filling

- Storage

14. Write a note on homogenizer and factors influencing homogenization ?

A **homogenizer** is a mechanical device that creates a stable and uniform mixture (an emulsion or suspension) by forcing a liquid through a narrow opening under high pressure. This process, known as homogenization, breaks down large particles or droplets (like the fat globules in milk) into much smaller, uniformly dispersed sizes, primarily through the physical principles of **turbulence, cavitation, and shearing**.

Factors Affecting Homogenization Efficiency

The effectiveness of the homogenization process can be influenced by several key factors:

- **Pressure**: This is a critical factor; higher pressure generally results in smaller particles and a more stable emulsion. Pressures for milk typically range from 100 to 250 bar (1500-3600 psi).
- **Temperature**: The product must be heated above the melting point of its fat (typically 55-80°C for milk) to ensure the fat is in a liquid state for proper subdivision. Homogenizing raw milk without prior heating can lead to rapid flavour deterioration due to enzyme (lipase) activity.
- **Product Composition (Fat/Protein Content)**: Products with high fat content (e.g., cream) are more difficult to homogenize efficiently because there may be insufficient protein available in the milk serum to form a new protective membrane around the increased surface area of small fat globules, leading to clumping.
- **Homogenizer Design/Configuration**:
  - **Valve type** (e.g., radial, axial flow) affects efficiency and flow characteristics.
  - **Single-stage vs. Two-stage**: Two-stage homogenization uses a second, lower-pressure stage to break up any fat globule clusters that form immediately after the first stage, improving overall stability.
- **Flow Rate/Processing Time**: Higher flow rates can result in lower efficiency if the particles do not spend enough time within the high-shear zone.
- **Viscosity**: The viscosity of the liquid affects how easily droplets can be deformed and broken apart by turbulent forces.

PART – C

Answer any 1 question

Each question carries 15 marks

15. How can you produce dahi from raw cow milk? Explain about the steps involved?

The milk must first be heat-treated to ensure safety and improve texture, and then inoculated with a specific bacterial starter culture under controlled temperature conditions.

Here are the steps involved:

- Receiving whole defatted milk
- Addition of skim milk powder
- Filtration
- Preheating
- Homogenization
- Pasteurization
- Cooling
- Inoculation
- Packaging
- Incubation
- Cooling and Storage

16. Write an essay on the factors affecting the composition of milk?

- Species
- Breed
- Individuality
- Interval of milking
- Day to day milking
- Age
- Frequency of milking
- Feeding
- Season
- Stage of lactation
- Completeness of milking